

**Lab #7: Control Design for an Inverted Pendulum**

**Released:** Mar 16, 2026

**Due:** Mar 18/19, 2026

## Objectives

By the end of this lab, you should be able to

1. Design a state-feedback controller for a linearized model of the inverted pendulum and use it to stabilize the pendulum in the upright position.
2. Understand the impact of control gains on the closed-loop system response and the trade-offs involved in control design.

**NOTE:** You do not need to solve things out by hand. Use numerical values and use MATLAB or Python on your own computer to compute everything needed to complete the lab.

## 1 Assigned Setup: Cart or Rotary?

The two setups are the cart-pole system (also called the single pendulum gantry) and the rotary pendulum system — you should work on the same setup that you worked on in Lab 5 and Lab 6.

The goal for both setups is the same — to use the linearized model (from labs 5 and 6) to design a state-feedback controller.

## 2 Background: Cart Pendulum

The cart pendulum system, also called the cart-pole system is a system where a cart moves in one dimension to control the angle of the pendulum. In this lab, we call this the linear single pendulum gantry (SPG) system. Note that this is not because it is a “linear system”, the linear here refers to the 1-D motion. A physical model of the cart-pole SPG system is shown in Figure 1.

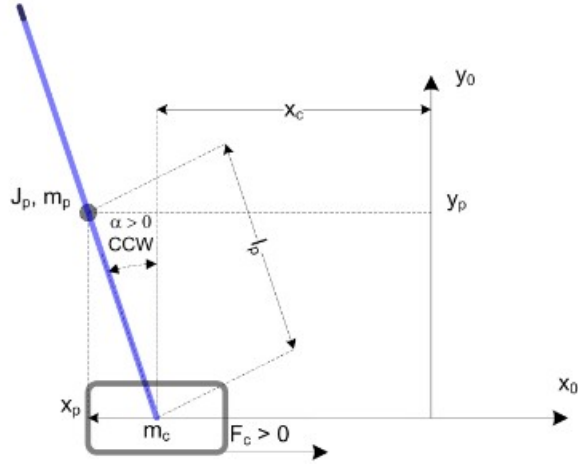


Figure 1: The cart-pole system (single pendulum gantry)

The pendulum pivot is on the IP02 cart, and is measured using the **Pendulum** encoder. The pendulum has mass,  $M_p$ , the centre of mass of the pendulum is at length,  $l_p$ , and the moment of inertia about the centre of mass is  $J_p$ . The pendulum angle,  $\alpha$ , is zero when it is suspended perfectly vertically and increases positively when rotated counter-clockwise (CCW). The positive direction of linear displacement of the cart,  $x_c$ , is to the right when facing the cart. The position of the pendulum centre of gravity is denoted as the  $(x_p, y_p)$  coordinate.

## 2.1 The linearized model of the cart-pole system

You have derived a linear model of the pendulum system by linearizing the nonlinear equations of motion for the cart and for the pendulum:

$$(J_{eq} + M_p)\ddot{x}_c + M_p l_p \cos(\alpha)\ddot{\alpha} - M_p l_p \sin(\alpha)\dot{\alpha}^2 = F_c - B_{eq}\dot{x}_c \quad (1)$$

and

$$M_p l_p \cos(\alpha)\ddot{x}_c + (J_p + M_p l_p^2)\ddot{\alpha} + M_p l_p g \sin(\alpha) = -B_p \dot{\alpha} \quad (2)$$

where  $F_c$  is the linear force applied to the cart,  $M_p$  is the mass of pendulum,  $l_p$  is the length of center of mass of pendulum from pivot,  $J_{eq}$  is the effective mass of the cart,  $J_p$  is the moment of inertia of the pendulum about the center of mass,  $g$  is the acceleration due to gravity, and  $B_{eq}$  is the equivalent viscous damping coefficient. This is our control input to the system.

### 3 Background: Rotary Pendulum

The rotary pendulum system uses a rotary arm (servo) to control the angle of the pendulum. The physical model of the rotary pendulum system is shown in Figure 2.

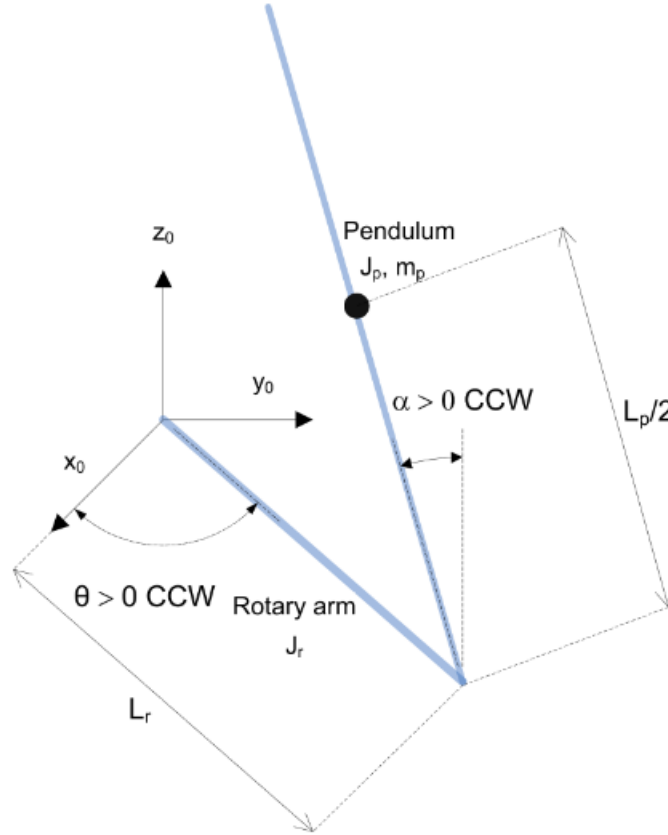


Figure 2: The rotary pendulum system

Starting from the two nonlinear equations of motion for the rotary arm and for the pendulum, you have derived the linearized state-space model. The nonlinear equations of motion, respectively, are:

$$\begin{aligned} & \left( m_p L_r^2 + \frac{1}{4} m_p L_p^2 - \frac{1}{4} m_p L_p^2 \cos(\alpha)^2 + J_r \right) \ddot{\theta} - \left( \frac{1}{2} m_p L_p L_r \cos(\alpha) \right) \ddot{\alpha} \\ & + \left( \frac{1}{2} m_p L_p^2 \sin(\alpha) \cos(\alpha) \right) \dot{\theta} \dot{\alpha} + \left( \frac{1}{2} m_p L_p L_r \sin(\alpha) \right) \dot{\alpha}^2 = \tau - B_r \dot{\theta} \end{aligned} \quad (3)$$

$$\begin{aligned}
& -\frac{1}{2}m_p L_p L_r \cos(\alpha) \ddot{\theta} + \left( J_p + \frac{1}{4}m_p L_p^2 \right) \ddot{\alpha} - \frac{1}{4}m_p L_p^2 \cos(\alpha) \sin(\alpha) \dot{\theta}^2 \\
& - \frac{1}{2}m_p L_p g \sin(\alpha) = -B_p \dot{\alpha}. \quad (4)
\end{aligned}$$

where  $\theta$  is the angle of the rotary arm,  $\alpha$  is the angle of the pendulum,  $m_p$  is the mass of the pendulum,  $L_r$  is the length of the rotary arm,  $L_p$  is the length of the pendulum,  $J_r$  is the moment of inertia of the rotary arm about its pivot,  $J_p$  is the moment of inertia of the pendulum about its center of mass,  $g$  is the acceleration due to gravity,  $B_r$  is the viscous damping coefficient of the rotary arm,  $B_p$  is the viscous damping coefficient of the pendulum, and  $\tau$  is the torque applied to the rotary arm. This is our control input to the system.

$$\tau = \frac{\eta_g K_g \eta_m k_t (V_m - K_g k_m \dot{\theta})}{R_m}. \quad (2.4)$$

## 4 Pre-lab work: state-space model

### Pre-Lab Quiz 4.1: Make sure you're ready for the lab!

Write the state-space model of the cart-pole system in the form  $\dot{x} = Ax + Bu$  and  $y = Cx + Du$ . Define the state vector  $x$ , the output vector  $y$ , and write down the matrices  $A$ ,  $B$ ,  $C$ , and  $D$  numerically. You only need to write the state-space model for the cart-pole system OR for the rotary pendulum system, depending on which setup you are working on.

Hint: Refer to the previous labs if you are confused about the definitions of the states and the outputs. If you don't have the matrices from the previous labs or if you're not sure, please find another team who may have completed and work together to quickly finish this pre-lab work.

## 5 Computational Steps: Control Design

Before starting the control design, describe the stability behavior of the system in open-loop.

### Pre-Lab Quiz 5.1: Make sure you're ready for the lab!

Compute the eigenvalues of the  $A$  matrix and determine whether the system is open-loop stable or unstable. Note down the eigenvalues and the stability conclusion in your answer.

You should see that your system is open-loop unstable, which means that the pendulum will not stay in the upright position without control action. The goal of this lab is to design a state-feedback controller that stabilizes the system in the upright position. To do this, we will first diagonalize the system by finding a similarity transformation that transforms the  $A$  matrix into a diagonal matrix. Then, we will design a state feedback controller in the diagonalized coordinates, and finally transform the controller back to the original coordinates.

### Pre-Lab Quiz 5.2: Make sure you're ready for the lab!

Obtain the diagonalized system by finding the similarity transformation that diagonalizes the  $A$  matrix. Write down the new state-space representation in the diagonalized coordinates.

**Hint:** Define the transformation  $z = Tx$ , where  $T$  is the matrix of eigenvectors of  $A$ . Then, the new state-space representation is given by  $\dot{z} = A_z z + B_z u$  and  $y = C_z z$ , where  $A_z = TAT^{-1}$ ,  $B_z = TB$ , and  $C_z = CT^{-1}$ .

Now that you have the diagonalized system, you can design a state feedback controller in the diagonalized coordinates. The first step is to choose where you would like your closed loop poles to be in the complex plane.

### Pre-Lab Quiz 5.3: Make sure you're ready for the lab!

Note down all open-loop pole locations again (from the previous quiz). Then, propose the new locations for the closed-loop poles that you would like to achieve with your state feedback controller. All chosen locations must be in the left half of the complex plane to ensure stability.

**Hint:** You may start with [TODO FILL STARTER VALUES TO HELP STUDENTS]

Now, let us try to find a controller that can achieve these closed loop poles. The control law is given by  $u = -\hat{K}z$ , where  $\hat{K}$  is the state feedback gain vector that you need to determine in the  $z$ -coordinate system. Use MATLAB or Python to find the values of  $\hat{K}$  that stabilize the system by using the `acker` or the `place` function. You

will need the matrices:  $A_z$ ,  $B_z$ , and the desired closed-loop pole locations (from the previous pre-lab quiz) to compute  $\hat{K}$ .

**Pre-Lab Quiz 5.4: Make sure you're ready for the lab!**

Compute the state feedback gain  $\hat{K}$  in the diagonalized coordinates using MATLAB or Python. Write down the values of  $\hat{K}$  that you obtained.

**Pre-Lab Quiz 5.5: Make sure you're ready for the lab!**

Compute the closed-loop matrix  $A_{cl,z} = A_z - B_z \hat{K}$  for the diagonalized system and compute its eigenvalues to verify that the closed-loop poles are at the desired locations. Use MATLAB to do this!

**Pre-Lab Quiz 5.6: Make sure you're ready for the lab!**

Obtain the controller gain in the original coordinates by transforming  $\hat{K}$  back to the original coordinates. The control law in the original coordinates is given by  $u = -Kx$ , where  $K = \hat{K}T$ . Write down the values of  $K$  that you obtained.

**Pre-Lab Quiz 5.7: Make sure you're ready for the lab!**

Prove that the system in the original coordinates, that is, the closed-loop matrix for your original state-space model  $A_{cl} = A - BK$ , has eigenvalues that are in the left half of the complex plane, which means that the system is stabilized by the control law  $u = -Kx$ .

## Pre-Lab Grade and Experimental Steps

Due: Mar 18/19, 2026 Name: \_\_\_\_\_

Grade: \_\_\_\_\_

# 1 Hands-on Experiment: Group Work

On the lab computer, make sure that you change the workspace directory to the folder on the Desktop labeled “EE 122” and under that “Lab #7”. Open the setup file for your assigned equipment and the associated Simulink model: `ip02_quick_start` for those working on cart-pole and for `q_rotpen_bal` for those working on the rotary pendulum.

0. Make sure to run `clear all` and `clc` to clear the workspace and the command window before starting the experiment so that your experiment is not affected by any previous work done on the computer and that you can reliably test your understanding.
1. Make sure that the values of the state-space matrices are correctly set in `SIP_ABCD_eqn_student.m` (for those on the cart-pole setup) or in the file named `ROTPEN_ABCD_eqns_student.m` (for those on the rotary pendulum setup). AFTER setting the matrix values, run the setup script.
2. Set the values of the controller gain  $K$  that you obtained by setting the variable `K` in the MATLAB workspace. Then, similar to previous labs, click “Build for Monitoring” on the Simulink model, then “Connect” and then “Start” to start the experiment.
3. To test whether the controller works, slowly bring the pendulum to an upright position and keep it there until you observe the controller take over. If the controller is working, you should see the pendulum stabilize at the upright position and the cart move to the desired position / rotary servo move to stabilize the pendulum. However, if the controller feels strange, don’t force it or leave it alone. Bring the pendulum down to zero and try again. If it still doesn’t work, stop the simulation and check your state-space matrices and gain values.
4. Open the cart position scope / rotary angle scope, the motor input voltage scope, and the pendulum angle scope to view the graphical data in real-time.

5. Identify which color trace corresponds to what variable in the scope.
6. Once you have observed the response, click “Stop” to stop the simulation.
7. Try another controller,  $K$ , by finding the controller gain for a different set of desired closed-loop pole locations and repeat the experiment.
8. Export the data from the MATLAB workspace variables to your personal computer to analyze the response further and to understand the impact of the modeling error perturbations you applied to the system in the previous step. Note that the data will be saved under the **Lab #7** folder in the **Lab\_7\_data** folder.

## 2 Action Items

In your lab report, provide the following:

1. Two controller gain matrices  $K_1$  and  $K_2$  that you designed for the system, along with the desired closed-loop pole locations that you chose for each controller.
2. Plots of the time response of the pendulum angle and the cart position (or rotary servo angle) for both controllers. Make sure to label your plots appropriately.
3. Using your data, compute the settling time and the percent overshoot for both controllers. Report the performance metrics and your reflections on the trade-offs between the two controllers in terms of performance and robustness.

| CL poles | $\hat{K}$ | $K$ | $t_s$ | PO |
|----------|-----------|-----|-------|----|
|          |           |     |       |    |
|          |           |     |       |    |
|          |           |     |       |    |